1 Chapter 2.3

Problem 13. *Give a combinatorial proof: If $n \geq 1$ and $k \geq 1$, then*

$$S(n,k) = \sum_{j=1}^{n} \binom{n-1}{j-1} S(n-j, k-1)$$

Answer: How many partitions can you make with an n-set and k-blocks?

Prior to starting this proof, I'll provide an example with a 5-set and 3 blocks. As a result, the written out formula appears as:

$$S(5,3) = \binom{4}{0} \cdot S(4,2) + \binom{4}{1} \cdot S(3,2) + \binom{4}{2} \cdot S(2,2) + \binom{4}{3} \cdot S(1,2) + \binom{4}{4} \cdot S(0,2)$$

For partitions, if $n < k$ then $S(n,k) = 0$. As such, the summation simplfies into.

$$S(5,3) = 1 \cdot S(4,2) + 4 \cdot S(3,2) + 6 \cdot S(2,2)$$

This is equivalent to taking one element, the number 1 in this example, and anchoring it to be inside a specific partition. At each iteration of j you can add another element to be inside the partition with the anchor value. See the table below for some examples.

j	1-Block	2-Block	3-Block
1	$\{1\}$	$\{2\}$	$\{3,4,5\}$
1	$\{1\}$	$\{3\}$	$\{2,4,5\}$
1	$\{1\}$	$\{4\}$	$\{2,3,5\}$
1	$\{1\}$	$\{5\}$	$\{2,3,4\}$
1	$\{1\}$	$\{2,3\}$	$\{4,5\}$
1	$\{1\}$	$\{2,4\}$	$\{3,5\}$
1	$\{1\}$	$\{2,5\}$	$\{3,4\}$
2	$\{1,2\}$	$\{3\}$	$\{4,5\}$
2	$\{1,2\}$	$\{4\}$	$\{3,5\}$
2	$\{1,2\}$	$\{5\}$	$\{3,4\}$
2	$\{1,3\}$	$\{2\}$	$\{4,5\}$
2	$\{1,3\}$	$\{4\}$	$\{2,5\}$
2	$\{1,3\}$	$\{5\}$	$\{2,4\}$
2	$\{1,4\}$	$\{2\}$	$\{3,5\}$
2	$\{1,4\}$	$\{3\}$	$\{2,5\}$
2	$\{1,4\}$	$\{5\}$	$\{2,4\}$
2	$\{1,5\}$	$\{2\}$	$\{3,4\}$
2	$\{1,5\}$	$\{3\}$	$\{2,4\}$
2	$\{1,5\}$	$\{4\}$	$\{2,3\}$
3	$\{1,2,3\}$	$\{4\}$	$\{5\}$
3	$\{1,2,4\}$	$\{3\}$	$\{5\}$
3	$\{1,2,5\}$	$\{3\}$	$\{4\}$
3	$\{1,3,4\}$	$\{2\}$	$\{5\}$
3	$\{1,3,5\}$	$\{2\}$	$\{4\}$
3	$\{1,4,5\}$	$\{2\}$	$\{3\}$

2 Chapter 2.4

Problem 1. *You have 40 pieces of candy to distribute among 10 children. Find the number of ways to do this in each of the following situations. Leave your answers in standard notation.*

(a) *The pieces of candy are different and each child gets at least one piece.*

(b) The pieces of candy are the same and each child get any number of pieces.

(c) The pieces of candy are different but you distribute them among 10 of the same paper bags.

(d) The candy is the same you distribute them in 10 bags that are the same and each bag contains at least one piece.

(e) The candy is different and each child gets exactly one pieces, so there is some left over candy.

(f) The candy is different and Frank receives four pieces.

Answer:

Problem 4. *Use type vectors to establish the bijection in Theorem 2.4.2.*

Theorem 1. *If $n \geq 1$ and $k \geq 1$, then:*

$$P(n,k) = \sum_{j=1}^{k} P(n-k, j)$$

Answer:

Problem 7. *Under what conditions on n and k is the statement $P(n,k) = P(n-1, k-1)$ true?*

Answer:

Problem 8. *Give a bijective proof of the following: The number of partitions of n is equal to the number of partitions of $2n$ into n parts.*

Answer: